

CS 3513 - Programming Languages

Programming Project (Group of 2 Students)

Problem Description

You are required to implement a lexical analyzer and a parser for the RPAL language. Refer RPAL_Lex.pdf for the lexical rules and RPAL_Grammar.pdf for the grammar details. You should not use 'lex', 'yacc' or any such tool.

Output of the parser should be the Abstract Syntax Tree (AST) for the given input program. Then you need to implement an algorithm to convert the Abstract Syntax Tree (AST) in to Standardize Tree (ST) and implement CSE machine.

Your program should be able to read an input file which contains a RPAL program. Output of your program should match the output of "rpal.exe" for the relevant program.

You must use C/C++ or Java for this project.

Input and Output Requirements

Your program should execute using the following

For c/c++:

\$./rpal20 file_name

For java:

\$ java rpal20 file_name

Where file_name is the name of the file that has the RPAL program as the input.

Input Format

Eg: Here is one input file

```
let Sum(A) = Psum (A,Order A )
           where rec Psum (T,N) = N eq 0 -> 0
                               | Psum(T,N-1)+T N
in Print ( Sum (1,2,3,4,5) )
```

Output Format

Output of the above program is:

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Submission

You must submit the following:

1. Makefile: Your make file must be directly under the zip folder. No nested directories.
2. Source Program: Provide comments.
3. REPORT:
 - The report should be in PDF format.
 - The report should contain your basic info: Name, and Student ID of both students.
 - Present function prototypes showing the structure of your programs. Include the structure of your program.

To submit, Please compress all your files together using a zip utility and submit to the Moodle System. Your submission should be named IndexNumber1_IndexNumber2.zip

All email submission will be ignored without further notification.

Executing the Program for C/C++:

You should create an executable file named rpal20. Provide a makefile to build your project. Make sure your program can be run with the following sequence.

```
> tar xvf <submission_file>.tar
> make
> ./rpal20 rpal_test_programs/rpal_01 > output.01
> diff output.01 rpal_test_programs/output01.test
> ./rpal20 rpal_test_programs/rpal_02 > output.02
> diff output.02 rpal_test_programs/output02.test
```

...

Executing the Program for java:

You should create rpal20.java as the main class. Provide a makefile to build you project. Make sure your program can be run with the following sequence.

```
> tar xvf <submission_file>.tar
> make
> java rpal20 rpal_test_programs/rpal_01 > output.01
> diff output.01 rpal_test_programs/output01.test
> java rpal20 rpal_test_programs/rpal_02 > output.02
> diff output.02 rpal_test_programs/output02.test
```

Grading Policy

Grading will be based on the correctness of algorithms. Below are some details of the grading

policy.

Correct implementation and execution: 70%

Comments and readability: 10%

Report: 20%

To grade your project, we will run your program on our test programs, and we will compare your output with the correct one. Full credit will be given a perfect match in every case. If your output does not match the correct output you will receive 0 marks for that test case.

Note: ***If you do not follow the input/output or submission requirements above, 25% of your score will be deduced.*** In addition, we may ask you to demonstrate your projects.